Complexity of Locally Injective k-Colourings of Planar Graphs

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Abstract

A colouring of the vertices of a graph is called injective if every two distinct vertices connected by a path of length 2 receive different colours, and it is called locally injective if it is an injective proper colouring. We show that for $k \geq 4$, deciding the existence of a locally injective k-colouring, and of an injective k-colouring, are NP-complete problems even when restricted to planar graphs. It is known that every planar graph of maximum degree $\leq \frac{3}{5}k - 52$ allows a locally injective k-colouring. To compare the behaviour of planar and general graphs we show that for general graphs, deciding the existence of a locally injective k-colouring remains NP-complete for graphs of maximum degree $2\sqrt{k}$ (when $k \geq 7$).

1. Definitions and Preliminaries

All graphs in this note are simple undirected graphs without loops. We write $N_G(v) = \{x \mid vx \in E(G)\}$ ($N_G[v] = \{v\} \cup N_G(v)$) to denote the open (closed, respectively) neighbourhood of a vertex $v \in V(G)$ in a graph G, and more generally $N_G[A] = \bigcup_{v \in A} N_G[v]$ and $N_G(A) = N_G[A] \setminus A$ for a set $A \subseteq V(G)$ of

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vertices of G. For an integer b, a graph is *b-bounded* if no vertex has more than b neighbours.

A proper k-colouring of a graph G is a map from the vertex set $V(G)$ of G to the set of k colours $[k] = \{1, \ldots k\}$ such that adjacent vertices are mapped to distinct colours. We refer to the decision problem of determining if an instance graph G has a k-colouring as k-COL.

A *locally injective k-colouring* of a graph G is a proper k-colouring χ that is injective when restricted to the neighbourhood $N_G[v]$, for every vertex v of G. We refer to the decision problem for locally injective k-colourings as k-LIHOM.

A *locally bijective k-colouring* is defined by replacing 'injective' with 'bijective'. The decision problem is k-LBHOM.

An *injective k-colouring* of a graph G is a vertex mapping $\chi : V(G) \to [k]$ which is not necessarily a proper k-colouring, but such that χ is injective on the neighbourhood $N_G(v)$, for every vertex v in G. The decision problem for injective k-colourings is referred to as k-INJ.

The notation is not chosen randomly. An edge preserving vertex mapping between two graphs is called a *homomorphism*, and homomorphisms are often called generalised colourings since a proper k-colouring of a graph is a homomorphism from it into the complete graph on k vertices. A homomorphism $f : G \to H$ is called *locally bijective* (*locally injective*) if for every vertex $v \in V(G)$, the restricted mapping $f : N_G[v] \to N_H[f(v)]$ is bijective (injective) [8]. It is easy to see that locally injective k-colourings are exactly locally injective homomorphisms into the complete graph K_k, and similarly for the locally bijective ones.

Locally bijective homomorphisms were intensively studied under the equivalent notion of graph covers [3, 4, 14, 19, 24]. Locally injective homomorphisms are closely related to distance constrained graph labelings [8], and in particular many papers on $L(2,1)$-labeling of graphs provide relevant results [5, 10, 15]. The computational complexity of deciding if an input graph allows a locally bijective (injective) homomorphism to a fixed target graph has been considered in [2, 4, 17, 18], but in neither of the two cases has a complete characteriza-

tion been obtained. Even less is known when the input graph is assumed to be planar. The restriction of graph colouring problems to planar inputs has been investigated in [22].

Another perspective is provided by distance powers of graphs. The d-th distance power of a graph G (usually denoted by $G^{(d)}$) is the graph with the same vertex set, and edges connecting every two vertices whose distance in G is at most d. It is straightforward to see that a locally injective k-colouring of a graph is a proper k-colouring of its second power, and vice versa. This approach can be found in [1, 7, 13].

Finally, injective colourings have been introduced rather recently in [11] and further results have been obtained in [6, 12, 21].

2. Background and Complexity Results

It is not hard to see that 3-LIHOM is polynomial time solvable. Indeed a graph G has a locally injective 3-colouring if and only if every component is a path or a cycle whose length is a multiple of 3.

In [8], it is shown that any instance of k-LBHOM can be reduced, in polynomial time, to an instance of k-LIHOM. Since it is shown in [16] that k-LBHOM is NP-complete for any $k \geq 4$, this shows that k-LIHOM is NP-complete for any $k \geq 4$.

We consider the complexity of k-LIHOM and k-INJ for planar graphs. While for $k \geq 4$, the NP-complete problem k-COL becomes polynomial time solvable when restricted to planar instances, the same does not happen for k-LIHOM. It is a consequence of [7] that 4-LIHOM is NP-complete for planar instances. Our first result, which we prove in Section 3, is to extend this to all $k \geq 5$.

Theorem 2.1. *For $k \geq 4$, k-LIHOM is NP-complete for planar instances.*

Since the only graphs that have locally injective k-colourings are $(k-1)$-bounded, Theorem 2.1 actually says that k-LIHOM is NP-complete for $(k-1)$-bounded planar graphs. It is shown in [23] that every $(\frac{3}{5}k - 52)$-bounded planar graph has a locally injective k-colouring. Further, it is conjectured by

Wegner that the same is true of every $(\frac{2}{3}k)$-bounded planar graph. It would be interesting to find the minimum b, $\frac{3}{5}k - 52 < b \leq k - 1$ for which k-LIHOM is NP-complete for b-bounded graphs.

Theorem 2.1 suggests that in terms of the complexity of k-LIHOM, restricting to planar instances has no effect. Our next result shows that this is not the case. It shows that compared to the non-planar problem, the planar problem is polynomial time solvable for graphs having much larger degrees. We prove the following in Section 5.

Theorem 2.2. *For $k \geq 7$, k-LIHOM is NP-complete when restricted to instances of maximum degree at most $2\sqrt{k-1}$.*

Finally, by providing a polynomial time reduction of any planar instance of k-LIHOM to a planar instance of k-INJ, we transport Theorem 2.1 to k-LIHOM. That is, we prove the following in Section 4.

Theorem 2.3. *For $k \geq 4$, k-INJ is NP-complete for planar instances.*

3. Proof of Theorem 2.1

The proof we give here of Theorem 2.1 for $k \geq 5$, can be slightly altered to give a constructive proof for the case $k = 4$. However, since this case follows from [7], we forgo the proof.

It was shown in [20] that the problem 3-COL is NP-complete even when restricted to planar instances. Using this result, we prove Theorem 2.1 as follows. For fixed k, we reduce any planar instance G of 3-COL, to a planar instance G' of k-LIHOM, in polynomial time. That is, we construct a planar graph G' such that G' has a locally injective k-colouring if and only if G has a 3-colouring. By the result of [20], this will prove the theorem.

The construction of G' is fairly complicated. In Subsections 3.1 to 3.6, we break it down into several smaller constructions, or gadgets. After constructing each gadget we observe certain properties, in the form of Lemmas 3.1 to 3.6. The proofs of these properties are all similar, and fairly straightforward, so we omit all but the first. After doing this, we formally begin the proof of Theorem 2.1 in Subsection 3.7.

We now define these gadgets for all $k \geq 5$.

3.1. The Doubling Gadget $D = D(k)$

Let $D = D(k)$ be as in Figure 1, where the dashed sets each contain $k-3$ vertices.

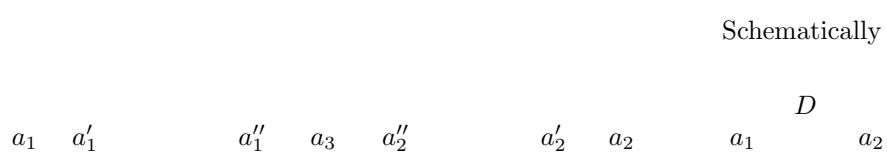

Figure 1: The Gadget $D = D(k)$

Lemma 3.1. *A mapping $\chi : \{a_1, a_2, a_1', a_2'\} \to [k]$ can be extended to a locally injective k-colouring of D if and only if the following conditions are satisfied.*

i. $\chi(a_1) = \chi(a_2)$
ii. $\chi(a_i) \neq \chi(a_i')$ *for* $i = 1, 2$

Proof. Suppose χ is a locally injective k-colouring of D. Since a_1' and a_1'' have degree $k-1$, their closed neighbourhoods contain all colours, each exactly once. Since $N_D[a_1'] \setminus \{a_1\} = N_D[a_1''] \setminus \{a_3\}$, it must be that $\chi(a_1) = \chi(a_3)$. Similarly, $\chi(a_3) = \chi(a_2)$. This proves (i), while (ii) follows immediately from the fact that χ is a proper colouring. On the other hand, a simple case analysis shows that any colouring of a_1, a_2, a_1', a_2' satisfying (i-ii) can be extended to a locally injective k-colouring of the whole gadget. □

Given an edge uv in a graph, we say that we *replace uv with D* to mean that we remove the edge uv, and then identify u with a_1 and v with a_2 of D.

3.2. The Switching Gadget $S = S(k)$

Let $S = S(k)$ be as in Figure 2 where the sets A and B each contain $k-3$ vertices. Here we use several copies of the gadget $D(k)$ from the previous subsection. These are represented by the schematic of $D = D(k)$ given in Figure 1.

Figure 2: The Gadget $S = S(k)$

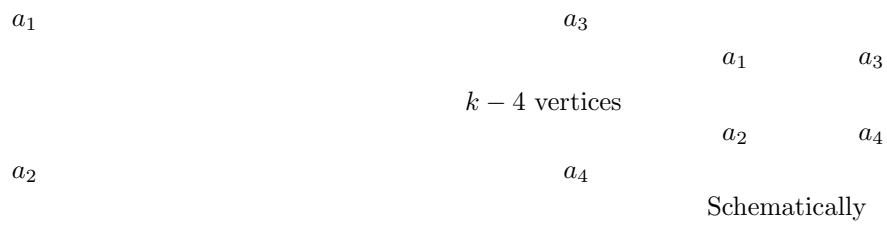

Figure 3: The Gadget $X = X(k)$

Lemma 3.2. *A mapping χ from $N_S[A \cup B]$ to $[k]$ can be extended to a locally injective k-colouring of S if and only if the following conditions are satisfied.*

i. *The sets $\{\chi(x) \mid x \in A\}$ and $\{\chi(x) \mid x \in B\}$ contain the same $k-3$ distinct colours.*
ii. *For $x \in A \cup B$, $\chi(x) \neq \chi(x')$ where x' is the unique neighbour of x in S.*

□

3.3. The Crossing Gadget X

Let $X = X(k)$ be as in Figure 3 where the dashed set contains $k-4$ vertices.

Lemma 3.3. *A mapping χ from $N_X[a_1, \ldots, a_4]$ to $[k]$ can be extended to a locally injective k-colouring of X if and only if the following conditions are satisfied.*

i. $\chi(a_1) = \chi(a_4) \neq \chi(a_3) = \chi(a_2)$

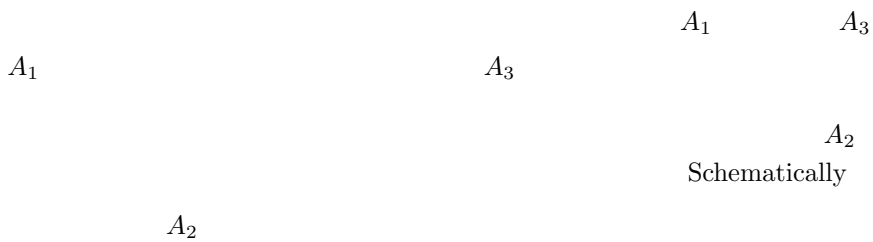

Figure 4: The Gadgets $Y(k)$.

ii. *For $x = a_1, \ldots, a_4$, $\chi(x) \neq \chi(x')$, where x' is the unique neighbour of x in X.*

□

3.4. The Splitting Gadget Y

Let $Y = Y(k)$ be the graph shown for $k = 7$ in Figure 4. For general k, the sets A_i for $i = 1, \ldots 3$ will each have $k - 3$ vertices. The graph Y is obtained from the three copies of the gadget $S = S(k)$ by drawing paths from the each vertex of B from one copy of S to a vertex of B in each of the other copies of S. Then each crossing is replaced by a gadget $X(k)$, and the two paths that do not cross any other paths are replaced by a gadget $D(k)$. (Hence the graph contains $(k-4) + (k-5) + \cdots + 1$ copies of $X(k)$.)

Lemma 3.4. *A mapping χ from $N_Y[A_1 \cup A_2 \cup A_3]$ to $[k]$ can be extended to a locally injective k-colouring of Y if and only if the following conditions are satisfied.*

i. *The sets $\{\chi(x) \mid x \in A_i\}$, contain the same $k - 3$ distinct colours for $i = 1, 2, 3$.*
ii. *For $x \in A_1 \cup A_2 \cup A_3$, $\chi(x) \neq \chi(x')$, where x' is the unique neighbour of x in Y.*

□

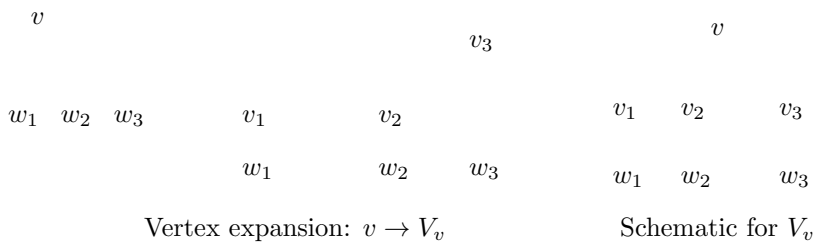

Figure 5: Vertex Expansion

Using the above gadgets, we will now define gadgets that we will use to replace the vertices and edges of a planar instance G of 3-COL, in order to reduce it to a planar instance of k-LIHOM.

3.5. Vertex Expansion $v \to V_v = V_v(k)$

Given integer $d \geq 2$, and $k \geq 5$, let $V_d(k)$ be a binary tree with d, in which we replace each edge with a copy of $D = D(k)$. Given a vertex v of degree d in some graph G, we say that we *expand* v, or *replace* v *with* V_v, to mean the following process. (See Figure 5).

- Remove v from G.
- Add a copy V_v of $V_d(k)$ to G and label the leaves $v_1, \ldots v_d$.
- Add a matching between the neighbourhood, $N_G(v)$, of v in G, and the set $\{v_1, \ldots, v_d\}$ of vertices of V_v.

Lemma 3.5. *A mapping χ from $N_{V_v}[v_1, \ldots, v_d]$ to $[k]$ can be extended to a locally injective k-colouring of V_v if and only if the following conditions are satisfied.*

i. $\chi(v_1) = \chi(v_2) = \cdots = \chi(v_d)$
ii. *For $x \in \{v_1, \ldots, v_d\}$, $\chi(x) \neq \chi(x')$, where x' is the unique neighbour of x in V_v.*

□

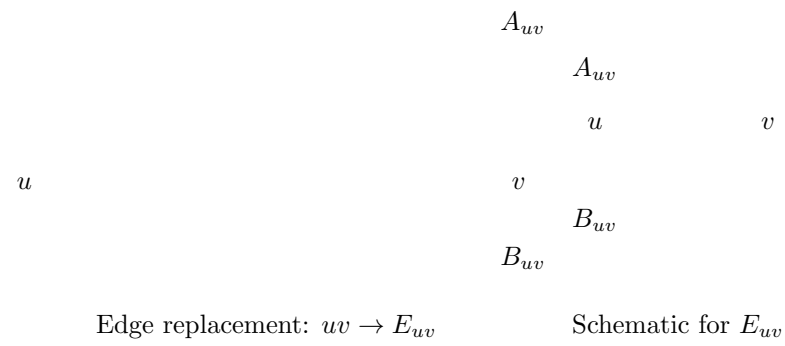

Figure 6: Edge replacements

3.6. Edge Replacement $e \to E_e = E_e(k)$

Given an edge $e = uv$ in some graph G, we say that we replace e (or uv) with E_e (or E_{uv}) to mean the following process.

- Remove the edge $e = uv$.
- Add the graph E_{uv} shown in Figure 6, between u and v.

Lemma 3.6. *A mapping χ from $N_{E_e}[A_e \cup B_e \cup \{u, v\}]$ to $[k]$ can be extended to a locally injective k-colouring of E_e if and only if the following conditions are satisfied.*

i. *The sets $\{\chi(x) \mid x \in A_e\}$ and $\{\chi(x) \mid x \in B_e\}$ contain the same $k - 3$ colours.*
ii. $\chi(u) \neq \chi(v)$
iii. $\chi(u), \chi(v) \notin \{\chi(x) \mid x \in A_e\}$
iv. *For $x \in A_e \cup B_e \cup \{u, v\}$, we have $\chi(x) \neq \chi(x')$, where x' is the unique neighbour of x in E_e.*

□

3.7. Proof of Theorem 2.1

Let G be a planar instance of 3-COL with a given embedding. We will construct a planar graph G' such that G has a 3-colouring if and only if G' has a locally injective k-colouring. We construct G' from G as follows. (See Figure 7 for an example of the construction.)

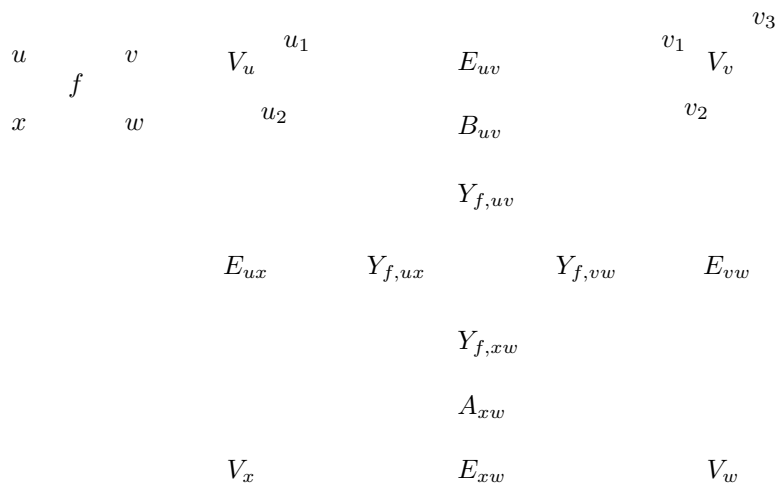

Figure 7: Construction: $G \to G'$.

- For each edge uv of G replace uv with E_{uv}.
- Expand each original vertex v of G. (Thus for every edge uv of G, the graph E_{uv} now contains the vertices u_i and v_j for some i and j.)
- For each edge e of each original face f of G, either A_e or B_e lies on f. For each such e and f let $Y_{f,e}$ be a copy of $Y(k)$ and identify the copy of A_2 in $Y_{f,e}$ with A_e or B_e, whichever lies on f. For consecutive edges e and e' of f, proceeding clockwise around the face, identify the copy of A_1 in $Y_{f,e}$ with the copy of A_3 in $Y_{f,e'}$. Do this preserving planarity.
- Call the resulting planar graph G'.

We now complete the proof by showing that G' has a locally injective k-colouring if and only if G has a 3-colouring. Let χ' be a locally injective k-colouring of G', and define the mapping $\chi : V(G) \to [k]$ by $\chi(v) = \chi'(v_1)$ for each $v \in V(G)$.

By the properties of $Y(k)$ and $E(k)$ (Lemmas 3.4 and 3.6), the sets $\{\chi'(x) \mid x \in A_e\} = \{\chi'(x) \mid x \in B_e\}$ are the same for all edges e of G. Assume,

without loss of generality, that this set is $\{4, \ldots, k\}$. Then by Lemma 3.6, $\chi'(v_1) \in \{1, 2, 3\}$ for all v in $V(G)$, so χ is actually a mapping of $V(G)$ to $\{1, 2, 3\}$. Furthermore, for each edge uv of G, there is an edge $u_i v_j$ in G' for some vertex u_i of V_u and v_j of V_v. Thus

$$\chi(u) = \chi'(u_1) = \chi'(u_i) \neq \chi'(v_j) = \chi'(v_1) = \chi(v),$$

and so χ is a 3-colouring of G.

Assume, for the inverse implication, that χ is a 3-colouring of G, and define a mapping χ' from $V(G')$ to $[k]$ as follows.

i. For each vertex v in G and for $i = 1, \ldots \deg(v)$, set $\chi'(v_i) = \chi(v)$.
ii. For each edge e of G, define χ' on A_e and B_e so that $\{\chi'(x) \mid x \in A_e\} = \{\chi'(x) \mid x \in B_e\} = \{4, \ldots, k\}$. Do the same for all sets A_1, A_2, and A_3 (in copies of $Y(k)$) on which χ' has not yet been defined.
iii. All vertices for which χ' has been defined so far have degree at most 3 and are distance at least 6 apart, so we can easily extend χ', as a locally injective k-colouring, to their neighbourhoods. Do this arbitrarily.

For any vertex x in G' for which $N_{G'}[x]$ is not yet fully coloured, $N_{G'}[x]$ is entirely within V_v or E_e for some vertex v or edge e of G, or within $Y_{f,e}$ for some face f of G and some edge e of f. Thus we need only extend χ' to a locally injective k-colouring of these gadgets.

For any edge uv of G, let u_i and v_j be the vertices of E_{uv} that are shared with V_u and V_v respectively. We have by step (ii) of our colouring that $\{\chi'(x) \mid x \in A_e\} = \{\chi'(x) \mid x \in B_e\} = \{4, \ldots, k\}$. This gives us condition (i) of Lemma 3.6. Step (i) gives us that $\chi'(u_i) = \chi(u) \neq \chi(v) = \chi'(v_j)$ are both in $\{1, 2, 3\}$, which is conditions (ii) and (iii) of Lemma 3.6. Step (iii) gives us condition (iv) of Lemma 3.6, and so χ' can be extended to a locally injective k-colouring of E_{uv}.

The verifications that χ' can be extended to the gadgets V_v and $Y_{f,e}$ are even more straightforward, so we omit them. This completes the proof of Theorem 2.1.

Figure 8: The Gadget $E_{uv}(k)$.

4. Proof of Theorem 2.3.

In this section, we reduce planar k-LIHOM to planar k-INJ, for $k \geq 4$, showing that planar k-INJ is NP-complete.

4.1. The Edge Gadget $E(k)$

Given an edge uv of some graph G, let $E_{uv}(k)$ be the graph in Figure 8, where the dashed set contains $k-2$ vertices.

Lemma 4.1. *A mapping χ of $\{u, v, u'_v, v'_u\}$ to $[k]$ can be extended to an injective k-colouring of $E_{uv}(k)$ if and only $\chi(u) = \chi(u'_v) \neq \chi(v) = \chi(v'_u)$.*

Proof. Suppose χ is an injective k-colouring of $E_{uv}(k)$. The $k-2$ vertices in the dashed set all have paths of length 2 between them, so they get distinct colours. All of the other vertices have paths of length 2 to each vertex of the dashed set, so each must get one of the remaining 2 colours. From the path u, v'_u, u', v', u'_v, v, it follows that $\chi(v) \neq \chi(v') \neq \chi(v'_u)$ and $\chi(u) \neq \chi(u') \neq \chi(u'_v)$ and thus $\chi(u) = \chi(v') = \chi(u'_v) \neq \chi(v) = \chi(u') = \chi(v'_u)$. □

4.2. Proof of Theorem 2.3.

Let G be a planar instance of k-LIHOM. Construct a planar graph G' from G by replacing each edge uv of with the graph $E_{uv} = E_{uv}(k)$ of Subsection 4.1. We now show that G has a locally injective k-colouring if and only if G' has an injective k-colouring. By Theorem 2.1, this will prove the theorem.

Let χ' be an injective k-colouring of G', and let χ be its restriction to $V(G)$. For any vertex v of G, Lemma 4.1 gives us that $\chi(v) = \chi'(v) \neq \chi'(u) = \chi(u)$ for any $u \in N_G(v)$. Moreover, χ' (and so χ) is injective on $N_G(v)$. Indeed if $\chi'(u_1) = \chi'(u_2)$ for $u_1 \neq u_2 \in N_G(v)$, then $\chi'(v'_{u_1}) = \chi'(u_1) = \chi'(u_2) = \chi'(v'_{u_2})$,

contradicting the fact that χ' is injective on $N_{G'}(v)$. Thus χ is a locally injective k-colouring of G.

On the other hand, let χ be a locally injective k-colouring of G, and define the mapping $\chi' : V(G') \to [k]$ as follows.

- For each vertex $v \in V(G) \subset V(G')$ let $\chi'(v) = \chi(v)$.
- For each edge uv of G, let $\chi'(u'_v) = \chi'(u)$ and and $\chi'(v'_u) = \chi(v)$ and extend χ' to an injective k-colouring of E_{uv}.

Since χ' is defined to be locally injective on copies of E_{uv} we just have to check that it is locally injective with respect to vertices whose neighbourhoods are not entirely within a single copy of E_{uv}. These are the vertices of G.

Let v be a vertex of G and assume that $\chi'(v'_{u_1}) = \chi'(v'_{u_2})$ for $v'_{u_1} \neq v'_{u_2} \in N_{G'}(v)$. Then $\chi(u_1) = \chi'(v'_{u_1}) = \chi'(v'_{u_2}) = \chi(u_2)$, contradicting the fact that χ is injective on $N_G(v)$. Thus χ' is a locally injective k-colouring of G'. This completes the proof of the theorem.

5. Proof of Theorem 2.2.

We show that for $k \geq 7$, k-LIHOM is NP-complete for $(2\sqrt{k-1})$-bounded graphs. We do this by taking any instance G of k-COL and reducing it to a $(2\sqrt{k-1})$-bounded instance G' of k-LIHOM. Since k-COL is NP-complete, this is enough.

For the proof we will construct a vertex replacement gadget $V = V(k)$, which depends on an auxiliary gadget $D = D(k)$.

5.1. Gadget $D = D(k)$

The gadget D varies slightly depending on k. There are two cases.

Case 1: $\ell^2 \leq k-1 \leq \ell(\ell+1)$ for some integer ℓ.

Let $r = (k-1) - \ell^2$, and define D as follows.

- Let $A = \{a_{i,j} \mid 1 \leq i,j \leq \ell\} \cup \{a_{\ell+1,j} \mid 1 \leq j \leq r\}$ be a set of $k-1$ independent vertices.

- Put an edge between $a_{i,j}$ and $a_{i',j'}$ if $i = i'$ or $j = j'$.
- Let $C = \{c_1, c_2, c_3\}$ be new vertices, and put an edge between $a_{i,j}$ and $c_{i'}$ if $i = i'$. (Since $k \geq 7$, $\ell \geq 3$, all vertices of C get neighbours in A.)

Lemma 5.1. *For $\ell^2 \leq k-1 \leq \ell(\ell+1)$, the following properties hold for D.*

i. *Under any locally injective k-colouring, the vertices of C all get the same colour.*
ii. *The vertices of A have degree at most $2\ell \leq 2\sqrt{k-1}$.*
iii. *The vertices of C have degree at most $\ell \leq \sqrt{k-1}$.*

Proof. For item (i) observe that the $k-1$ vertices of A are distance at most 2 apart, so all get distinct colours, and the vertices of C are distance 3 apart and distance at most 2 from everything in A. The rest of the proof is straightforward. □

Case 2: $\ell(\ell+1) < k-1 < (\ell+1)^2$ for some integer ℓ.

Let $r = (k-1) - \ell(\ell+1)$, and define D as follows.

- Let $A = \{a_{i,j} \mid 1 \leq i \leq \ell+1, 1 \leq j \leq \ell\} \cup \{a_{i,\ell+1} \mid (\ell+1)-(r-1) \leq i \leq \ell+1\}$ be a set of $k-1$ independent vertices.
- Put an edge between $a_{i,j}$ and $a_{i',j'}$ if $i = i'$ or $j = j'$.
- For $i' = (\ell+1)-(r-1), \ldots, \ell+1$, and $i = 1, \ldots 3$, if there is no vertex $a_{i,\ell+1}$, then put an edge between $a_{i',\ell+1}$ and $a_{i,i'-1}$.
- Let $C = \{c_1, c_2, c_3\}$ be a set of new vertices, and put an edge between $a_{i,j}$ and $c_{i'}$ if $i = i'$.

Lemma 5.2. *For $\ell(\ell+1) < k-1 < (\ell+1)^2$, the following properties hold for D.*

i. *Under any locally injective k-colouring, the vertices of C all get the same colour.*
ii. *The vertices of A have degree at most $2\ell+1 \leq 2\sqrt{k-1}$.*
iii. *The vertex c_1 has degree ℓ, and the other vertices of C have degree at most $\ell+1$.*

Proof. We only observe how $2\ell+1 \leq 2\sqrt{k-1}$ follows from the condition $\ell(\ell+1) < k-1$. Indeed, by completing the square one gets $\ell + 1/2 < \sqrt{k-3/4}$, so $2\ell+1 < \sqrt{4k-3}$. Now if $2\ell+1 > 2\sqrt{k-1} = \sqrt{4k-4}$, then $\sqrt{4k-4} < 2\ell+1 < \sqrt{4k-3}$, and so there is a proper square strictly between $4k-4$ and $4k-3$, which is impossible. □

5.2. Vertex replacement $v \to V_v = V_v(k)$.

Let $k \geq 7$ be fixed. Given an integer $d \geq 2$, let V_d be the graph constructed from d mutually vertex disjoint copies $D_1, \ldots, D_d$ of the graph $D(k)$ by identifying the copy of c_3 in D_i and the copy of c_1 in D_{i+1} for $i = 1, \ldots, d-1$.

Given a vertex v of degree d in a graph G, we say that we *expand* v or *replace* v *with* V_v to mean the following process.

- Remove v from G, and add d edgeless vertices $v_1, \ldots, v_d$.
- Add a copy V_v of V_d to G, for $i = 1, \ldots, d$, identify the copy of c_2 in D_i in V_v with v_i.
- Add a matching between $\{v_1, \ldots, v_d\}$ and $N_G(v)$.

Lemma 5.3. *The following properties hold for V_v.*

i. *For any locally injective k-colouring χ, $\chi(v_1) = \cdots = \chi(v_d)$.*
ii. *The maximum degree in V_v is $2\sqrt{k-1}$.*
iii. *The maximum degree in V_v of any vertex of V_v with neighbours outside of V_v (i.e., any copy of c_2) is at most $\sqrt{k-1}+1 < 2\sqrt{k-1}-1$.*
iv. *For any choice of colours $x_0, x_1, \ldots, x_d$ from $[k]$ with $x_0 \neq x_i$ for $i = 1, \ldots, d$, there is a locally injective k-colouring χ of V_v such that $\chi(v_1) = \cdots = \chi(v_d) = x_0$, and for $i = 1, \ldots, d$, no neighbour of v_i in V_v gets colour x_i, i.e., (*)*

$$x_i \notin \{\chi(u) \mid u \in N_{V_v}(v_i)\}.$$

Proof. The first three properties follow easily from Lemmas 5.1 and 5.2. To see property (iv) for Case 1, define χ as follows. Set χ to x_0 on all vertices of C in all copies of $D(k)$ in V_v. For $i = 1, \ldots, \ell$, set χ to x_i on the copy of $a_{1,1}$ in D_i. This gives us (*). It is easy now to extend χ to a locally injective k-colouring of V_v. We define χ greedily, beginning on D_1, and proceeding to D_2 and continuing through to D_ℓ. We must only be careful that when we define χ on D_i we let $\chi(a_{1j}) = x_{i+1}$ for some j, so that it is distance 3 from the vertex a_{11} of D_{i+1}. The proof in Case 2 is similar. □

5.3. Proof of Theorem 2.2.

Let G be an instance of k-COL, and let G' be the graph constructed by expanding each vertex of G. By Lemma 5.3 the maximum degree of G' is at most $2\sqrt{k-1}$. Thus, using the fact that k-COL is NP-complete, we prove the theorem by showing that G has a k-colouring if and only if G' has a locally injective k-colouring.

Let χ' be a locally injective k-colouring of G', and define $\chi : V(G) \to [k]$ by setting $\chi(v) = \chi'(v_1)$ for all $v \in V(G)$. Then for every edge $uv \in G$, there exists an edge $u_i v_j$ in G' for some $u_i \in V_u$ and $v_j \in V_v$, so

$$\chi(u) = \chi'(u_1) = \chi'(u_i) \neq \chi'(v_j) = \chi'(v_1) = \chi(v).$$

Thus χ is a k-colouring of G.

On the other hand, let χ be a k-colouring of G, and define $\chi' : V(G') \to [k]$ as follows. For $v \in V(G)$, property (iv) of Lemma 5.3 implies that there is a locally injective k-colouring χ'_v of V_v such that $\chi'_v(v_1) = \cdots = \chi'_v(v_{d_v})$ where $d_v = |N_G(v)|$, and for each $w \in N_G(v)$, $\chi(w) \notin \{\chi'_v(u) \mid u \in N_{V_v}(v_i)\}$, where v_i is the neighbour of w in V_v. Let χ' restricted to $V(V_v)$ be this χ'_v.

By its definition via the χ'_v, χ' is locally injective on any vertex whose neighbourhood is entirely within V_v for some $v \in V(G)$. We just have to check that it is locally injective with respect to v_i for $v \in V(G)$ and $i = 1, \ldots, |N_G(v)|$. But again by definition of χ' the neighbours of v_i in V_v get distinct colours that are different from $\chi(w)$, which is the colour of the only other neighbour of v_i in G'.

Thus χ' is a locally injective k-colouring of G'. This completes the proof of the theorem.

6. Conclusion and Final Remarks

We have shown that for every $k \geq 4$ deciding the existence of a locally injective k-colouring is NP-complete for planar inputs. This can be viewed as a start to discussing the computational complexity of the question of locally injective homomorphisms into fixed graphs for planar inputs. It is plausible to conjecture that for every fixed target graph H, the problem restricted to planar inputs is as difficult as the general one. However, if this is true, it may not be so easy to prove, since it would imply that $L(2,1)$-labeling of span 4 is NP-complete for planar graphs, which is a well known open problem in the area of Frequency Assignment Problems (known to be hard only for span 8 [5]).

Another natural question is the complexity of locally bijective homomorphisms restricted to planar inputs. Here the first open question is already posed by K_4 as the target graph. Or equivalently, how difficult is to decide existence of locally injective 4-colouring for planar cubic graphs? (The result of [7] concerns only subcubic graphs, i.e., graphs of maximum degree 3).

We have further shown that for general graphs, locally injective k-colouring remains NP-complete for graphs of substantially smaller maximum degree (for $(2\sqrt{k-1})$-bounded graphs). This is (upto a multiplicative constant) best possible, since every $(\sqrt{k-1})$-bounded graph is locally injective k-colourable. However, any improvement on the bound of maximum degree would be interesting even for small values of k. For instance, the construction in the proof of our Theorem 2.2 works for $k \geq 7$. So what is the smallest d such that 6-LIHOM is NP-complete for d-bounded graphs? Our current best bound is 5.

In case of planar inputs, our constructions yield graphs that have maximum degree $k-1$. It would be interesting to know if this bound can be improved.

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